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grouper: An R Package for Optimal Group Assignment in Higher Education

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Software

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Summary

Collaborative learning pedagogies are increasing in prominence in higher education. However, the realisation of these pedagogies' benefits depend on how educators assign learners to groups. Because such assignments can be time-consuming and subjective, there is a need for an algorithmic solution for them. In this paper, we present `grouper`, an easy-to-use and free R package that implements two flexible group allocation strategies in an automated and objective manner. `grouper` was designed to ease the burden on educators adopting collaborative learning pedagogies in their courses, and has been evaluated on two undergraduate courses described here.

Statement of Need

Institutes of Higher Learning (IHLs) are increasingly using collaborative learning pedagogies, which can benefit learners through deeper understanding of course content and teamwork skills. However, the realisation of these sought-after benefits depend on how educators assign learners to groups.

Educators have formulated various mathematical models to perform this assignment: (Anwar & Bahaj, 2003) and (Meyer, 2009) developed models that prioritised maximising students' project preferences; (McGuirk, 2020) developed a model that prioritised students' preferences, group sizes and group composition. Other models address related, but distinct, problems such as assigning students to elective courses (Beroš & Meter, 2015) or incorporating staff workload into student-to-project supervisor assignments (Ramotsisi et al., 2022).

As part of our research, we also conducted in-depth interviews with two professors from contrasting disciplines.

The first professor, from the School of Computing at National University of Singapore, allowed students to self-form sub-groups of up to three members in his course. These sub-groups were then paired to create larger project teams of six, in order to neutralize differences in skill levels across the final teams.

We utilised a similar procedure in a course in our department (Department of Statistics and Data Science). However, instead of utilising skills, we prioritised each self-formed sub-group's preference for their roles and topics. While self-formed sub-groups ensure familiarity between members in the sub-group, matching sub-groups randomly possibly forces students to collaborate with others whom they may not know well, replicating workplace scenarios.

The second professor, from the Department of Communications and New Media, provided additional perspective on practical challenges in group allocation for courses in the Arts

and Social Sciences. For her, the main objective in group allocation is to achieve diversity, especially for certain topics/tasks; the ideal group should encompass varied viewpoints and expertise, drawn from different academic specializations and personal backgrounds, e.g. year of study. She explained that having group members from diverse backgrounds facilitates understanding the problem from multiple angles, generally leading to more innovative solutions.

In summary, our literature review and interviews highlight several factors that educators could prioritise when performing group assignment in higher education. Whichever factor is used, it is apparent that there is a need for an algorithmic solution for the assignment. This would ease the burden on the instructor, while providing an objective procedure for the assignment. Our contribution is an R package `grouper` that offers two flexible group allocation strategies.

Functionality and Usage

`grouper` provides two distinct integer linear programming optimisation models.

The Preference-Based Assignment (PBA) model allows educators to assign student groups to topics to maximise overall student preferences for those topics. The topics can be viewed as project titles. The model allows for repetitions of each project title. This formulation also allows each project team to comprise multiple sub-groups.

To execute the optimisation routine, an instructor prepares:

1. A group composition table listing the member students within each self-formed group.
2. A preference matrix containing the preference that each self-formed group has for each topic.
3. A YAML file defining the remaining parameters of the model.

The Diversity-Based Assignment (DBA) model enables educators to assign students to groups and topics with the dual, but weighted, aims of maximising diversity (based on student attributes) within groups and balancing specific skill levels across different groups.

To execute the DBA optimisation routine, the instructor prepares:

1. A group composition table containing:
   - the member students within each self-formed group,
   - the demographics that will be used to compute pairwise dissimilarity between students, and
   - a numeric measure of each student's skill.
2. A YAML file defining the remaining parameters of the model.

Application to Courses

To Interdisciplinary Course

The final author of this paper teaches an interdisciplinary course together with a colleague from the English Language department. Students first learn about data science and linguistics in lectures. Then, in small groups, they discuss chosen topics from the viewpoint of linguistics and data science/statistics. As such, it is necessary to create groups with a good balance of Data Science and Statistics (dsds) majors, and non-dsds majors.

The table below shows a manual breakdown of 24 students in a tutorial class, into 5 discussion groups. The left two columns describe the result of manual assignment. And the two right-most columns describe the result of assignment using the DBA model in `grouper`.

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	manual_assigned		grouper_assigned	
group_id	dsds	non-dsds	dsds	non-dsds
1	2	3	3	2
2	4	1	3	2
3	3	2	2	3
4	2	3	2	2
5	2	2	3	2

It is clear that the routine does a fairer assignment. Note that, as there are 5 such tutorial groupings, this assignment has to be carried out five times. Here is a visualisation of the groupings:

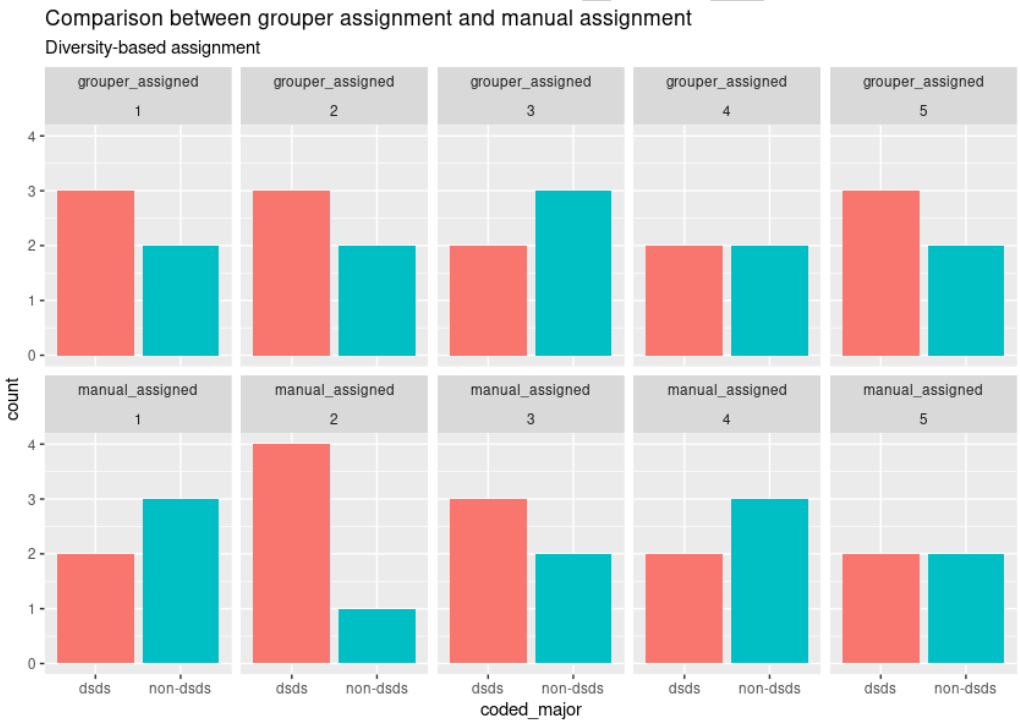

Figure 1: Diverse groupings

To Course with Sub-group Structure

In the "Statement of Need" section, we mentioned a Data Science course that requires allocating students into project teams with 2 sub-groups: one sub-group typically develops a user-interface; and the other works on data analysis. The table below shows how `grouper` could have benefited us if it had been used over the four semesters the course was taught.

Consider the first semester we ran the course (row 1): the enrolment was 62 and there were 4 topic titles to be allocated (with some permitted repetitions). In the manual allocation carried out, the average preference of the self-formed groups was 8.16. *If we could have used* `grouper`, the mean preference for this same group of students would have been 9.47 (higher is better).

Sem	manual_pref	grouper_pref	class_size	n_topics	n_project_teams
1	8.16	9.47	62	4	7

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Sem	manual_pref	grouper_pref	class_size	n_topics	n_project_teams
2	13.10	13.30	108	7	13
3	12.70	13.50	89	7	11
4	9.48	9.66	151	5	18

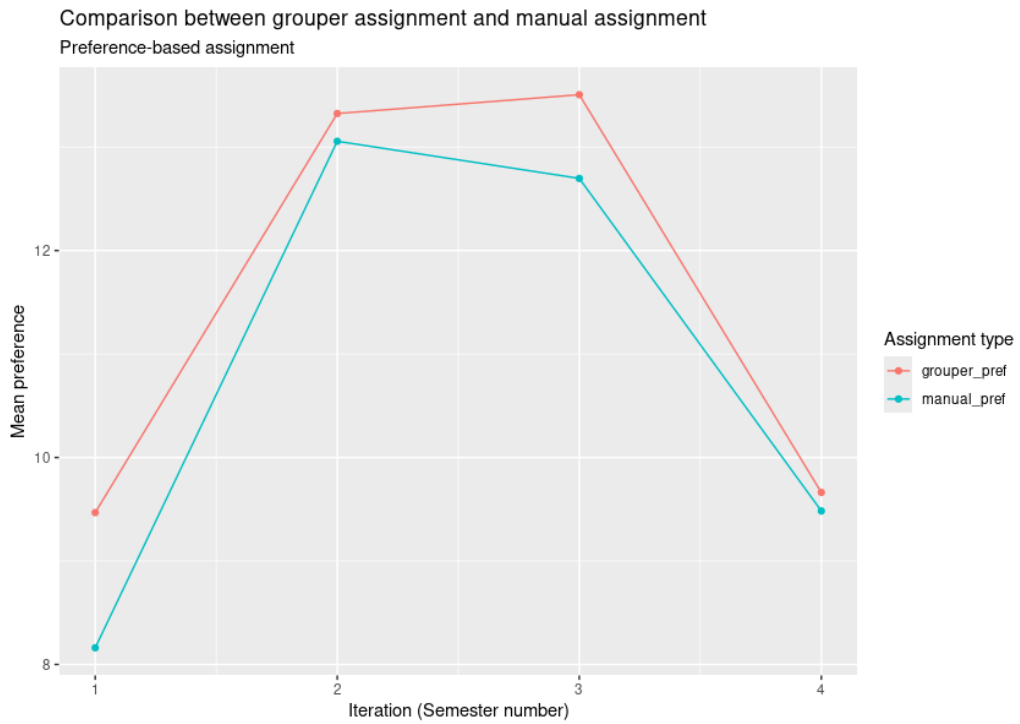

Figure 2: Preference-based groupings

It is clear that `grouper` would have improved the overall preference if we had access to it.

Discussion

While it seems as though the improvement proffered in both applications above is minimal, the reader should note that manual allocation is extremely arduous. For the data science course, manual assignment typically takes a week or more. With `grouper`, it only takes 2 minutes! Using it allows the educator time to focus more on course content and student guidance.

Ethical Approvals

This study has been cleared by National University of Singapore IRB (ref. NUS-IRB-2025-547). This study was not funded. The authors have no conflicts of interest to declare.

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